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Apolipoprotein AI_{Milano}

ACCELERATED BINDING AND DISSOCIATION FROM LIPIDS OF A HUMAN APOLIPOPROTEIN VARIANT*

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The lipid binding properties of apolipoprotein (apo) AI_{Milano}, a molecular variant of human apolipoprotein AI, characterized by the $Arg^{173} \rightarrow$ Cys substitution, was investigated by the use of dimyristoylphosphatidylcholine liposomes. Both the variant AI_{Milano} and normal AI are incorporated to the same extent in stable complexes isolated by gel filtration, showing similar dimensions and stoichiometries. A higher affinity of apo-AI_{Milano} for dimyristoylphosphatidylcholine is suggested by the faster association rate of the variant apoprotein compared to normal AI; similarly, apo-AI_{Milano} is more readily displaced by guanidine hydrochloride from the isolated dimyristoylphosphatidylcholine-apoprotein complexes. When the secondary structure of apo-AI_{Milano} was investigated by spectrofluoroscopy and circular dichroism, a higher fluorescence peak wavelength and a lower α-helical content were detected in the variant apoprotein compared to normal AI. The substitution $Arg^{173} \rightarrow$ Cys in the AI_{Milano} dramatically alters the amphipathic nature of the modified α-helical fragment of apoprotein AI. The association rate with lipids is accelerated by an increased exposure of hydrophobic residues. The reduced stability of the lipid-apoprotein particles is possibly mediated by a reduction in the number of helical segments involved in lipid association. The high flexibility of the AI_{Milano} apolipoprotein in the interaction with lipids may explain its accelerated catabolism and the possibly improved uptake capacities for tissue lipids.

Human apolipoprotein AI is the major protein constituent of plasma high density lipoproteins. HDL[1] play a critical role in the process of cholesterol transport from tissues to plasma and a significant correlation has been detected in clinical and epidemiological studies between low plasma apo-AI and HDL cholesterol and increased risk of premature cardiovascular disease (1, 2). Apo-AI is a single polypeptide chain, composed of 243 amino acids (3), and can interact with different phosphatidylcholines to form stable complexes of the general size of HDL (4, 5). This ability of apo-AI to spontaneously associate with phospholipids is a phenomenon common to other apolipoproteins (AII, CI, CII, CIII, E) and is attributed to their potential to form amphipathic α-helices (6). The amphipathic structure arises when apolar and polar residues in the amino acid sequence of the apolipoprotein are located on opposite sides of the α-helix (7). This amphipathic nature of the helix allows the apolipoprotein to penetrate into a lipid bilayer, leading to its disruption.

DMPC and apo-AI have been long used as a model system for investigating the lipid-protein interactions in HDL (8, 9). Apo-AI spontaneously interacts with aqueous dispersions of DMPC to form stable complexes, separable by gel filtration and/or gradient ultracentrifugation (4, 10). The extent of the interaction is kinetically controlled (11), the rate of association being increased by the presence of cholesterol (12) and being fastest at the gel to liquid-crystalline phase transition temperature of the lipid (13). The binding of apo-AI to DMPC is a reversible process, apo-AI being dissociated from isolated complexes by heating (9) or by treatment with denaturing agents such as GdnHCl (14). The stoichiometry and the size and shape of the resulting complexes critically depend upon the initial proportions of DMPC and apo-AI in the incubation mixture (4, 10). The small micellar complexes obtained at low DMPC/apo-AI ratios (100:1) are similar to "nascent" HDL synthesized in the liver and gut (15, 16) and/or to plasma HDL of patients with deficiency of lecithin cholesterol acyltransferase (17). These complexes are effective substrates for the enzyme, resulting in the conversion of discoidal complexes to spherical particles, similar to circulating HDL (18).

The apolipoprotein AI_{Milano} has been the first described molecular variant of human apolipoproteins (19, 20); it is characterized by a cysteine for arginine replacement at position 173 in the amino acid sequence (21). The apo-AI_{Milano} carriers have markedly low HDL-cholesterol levels but also show a very low prevalence of atherosclerotic disease (22). HDL, isolated from these subjects show an abnormal lipid composition (19). The HDL_2 subfraction is nearly absent, whereas HDL_3 are extremely heterogeneous when compared to lipoproteins similarly isolated from normal subjects (23).

In the present study, the effect of the single Arg $\rightarrow$ Cys substitution in the apo-AI_{Milano} on the structural and lipid binding properties of apo-AI, was investigated. The reported findings suggest that this amino acid replacement may be responsible, by way of a slight modification in the secondary structure of apo-AI, of some of the characteristic biochemical features detected in the apo-AI_{Milano} carriers and, possibly, of a more efficient removal of tissue lipids.

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[1] The abbreviations used are: HDL, high density lipoproteins; apo, apolipoprotein; DMPC, dimyristoylphosphatidylcholine; AI_M, AI_{Milano}; GdnHCl, guanidine hydrochloride; SDS, sodium dodecyl sulfate.

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EXPERIMENTAL PROCEDURES

Preparation of Apoproteins

Plasma HDL (ρ = 1.063–1.21) were isolated from a blood donor and from different apo-AI_{Milano} affected subjects by sequential ultracentrifugation in a 50 Ti rotor (Beckman, Palo Alto, CA). HDL were extensively dialyzed against 5 mM NH_4HCO_3 containing 0.01% EDTA (pH 7.4), lyophilized, and delipidated with $CHCl_3$/MeOH (2:1, v/v). The apo-HDL were solubilized in 0.1 M Tris-HCl, 0.04% EDTA, 0.01% NaN_3 containing 6 M GdnHCl and 1% 2-mercaptoethanol (pH 7.4). After 4 h of incubation at 37 °C, reduced apo-HDL were applied to a Sephacryl S-200 (Pharmacia, Uppsala, Sweden) column (2.6 × 150 cm) equilibrated with 0.1 M Tris-HCl, 0.04% EDTA, 0.01% NaN_3 containing 4 M GdnHCl and 0.1% 2-mercaptoethanol (pH 7.4). Pooled apo-AI fractions were reapplied to the same column, dialyzed against 5 mM NH_4HCO_3 (pH 7.4), and lyophilized. Apo-AI from both normal and AI_{Milano} subjects gave a single band in SDS-polyacrylamide gel electrophoresis when 100 μg of protein were applied to a 1.5-mm thick slab gel.

Apo-AI_{Milano} was separated from total apo-AI, isolated from carriers, by affinity chromatography through thiopropyl-Sepharose 6B (Pharmacia, Uppsala, Sweden) as previously described (21). Purified apo-AI_{Milano} differs from normal apo-AI for the lack of the major AI isoprotein, as assessed by analytical isoelectric focusing (21) and for the presence of 1 cysteine residue for mole of protein as demonstrated by the quantitation of sulfhydryl groups with 5,5′-dithiobis-(2-nitrobenzoic) acid (24).

Recombination Procedures

Weighed amounts of DMPC were dissolved in ethanol and the solvent evaporated under N_2; any remaining solvent was removed under vacuum for 2 h. Dispersions of DMPC in 0.01 M Tris-HCl, 1 mM EDTA (pH 7.4) containing 8.5% KBr (4), were mixed with both apo-AI_{Milano} and apo-AI at 100:1 DMPC/AI molar ratio. After incubation for 20 h at 25 °C, the complexes were isolated by chromatography through a calibrated Bio-Gel A-5m (Bio-Rad, Richmond) column (1.5 × 90 cm), equilibrated with 0.01 M Tris-HCl, 0.15 M NaCl, 1 mM EDTA (pH 7.4). Pooled fractions were concentrated with an Amicon cell, equipped with PM10 membranes. Individual fractions and concentrated complexes were analyzed for protein content by the Lowry method (25) and for phospholipids by an enzymatic assay (26). The apolipoprotein and DMPC yields were constantly greater than 93% for the entire series of recombination experiments.

The kinetics of association between apoproteins and DMPC were followed by recording the decrease of absorbance at 325 nm in a Gilford model Stasar III spectrophotometer, equipped with a thermostatically controlled cuvette, maintained at 25 °C (13).

The stability of purified apoproteins and of their complexes with DMPC was assessed by recording the tryptophanyl fluorescence spectra after a 4-h incubation at 25 °C with GdnHCl at concentrations varying between 0 and 8 M (27).

Spectroscopy—Uncorrected, intrinsic protein fluorescence spectra were recorded on a Jasco FP-550 spectrofluorometer at 26 °C using an excitation of 280 nm. Circular dichroism spectra were recorded with a Jasco J500A spectropolarimeter at 25 °C. Mean residue ellipticity values $|\vartheta|$ were expressed in degrees × cm^2 × $dmol^{-1}$ and were calculated by the equation:

$$|\vartheta| = \frac{\vartheta \times 106}{10 \times l \times c}$$

where ϑ is the observed ellipticity in degrees, 106 the mean residue molecular weight of the proteins (3), l the path length in cm, and c the protein concentration in g/ml.

Analyses of the secondary structural information, contained in the far-UV spectra, were carried out by the method described by Yang and co-workers (28), who kindly provided us with the circular dichroism data of 16 reference proteins of known secondary structure. These were used for computing the experimental values of our samples. The calculations were performed on a Gould S.E.L. 32/7780 computer.

Other Methods—Analytical isoelectric focusing and SDS-polyacrylamide gel electrophoresis were performed as previously described (29). The statistical analyses were carried out by standard methods (30).

RESULTS

Isolation and Characterization of the Complexes of Normal Apo-AI and Apo-AI_{Milano} with Dimyristoylphosphatidylcholine—The same preparations of normal AI from a blood donor and of pooled monomeric apo-AI_{Milano} from three carriers have been used in all the recombination studies. Typical column fractionations by gel filtration of mixtures consisting of DMPC and apo-AI or apo-AI_{Milano} are shown in Fig. 1. The elution profiles from the agarose column reveal the presence of a lipid-protein complex eluting just after HDL. Variable amounts of unreacted DMPC and apoprotein elute at the void volume and near the total column volume, respectively. The percentage of apo-AI_{Milano} incorporated in the complex is slightly lower than that of normal apo-AI (90.6 ± 4.2 *versus* 93.4 ± 3.1, mean ± S.D. of four different experiments). However, the nearly identical elution volumes and stoichiometries indicate that similar complexes are formed with both apoproteins (Table I).

The association with DMPC induced a shift toward the blue of the fluorescence wavelength maxima of tryptophanyl residues in both apoproteins: in apo-AI from 335.2 to 333.0 nm, in apo-AI_{Milano} from 337.4 to 334.7 nm. The fluorescence intensity at the maxima increased around 30% (Table I). The circular dichroism spectra showed a rise of the α-helical

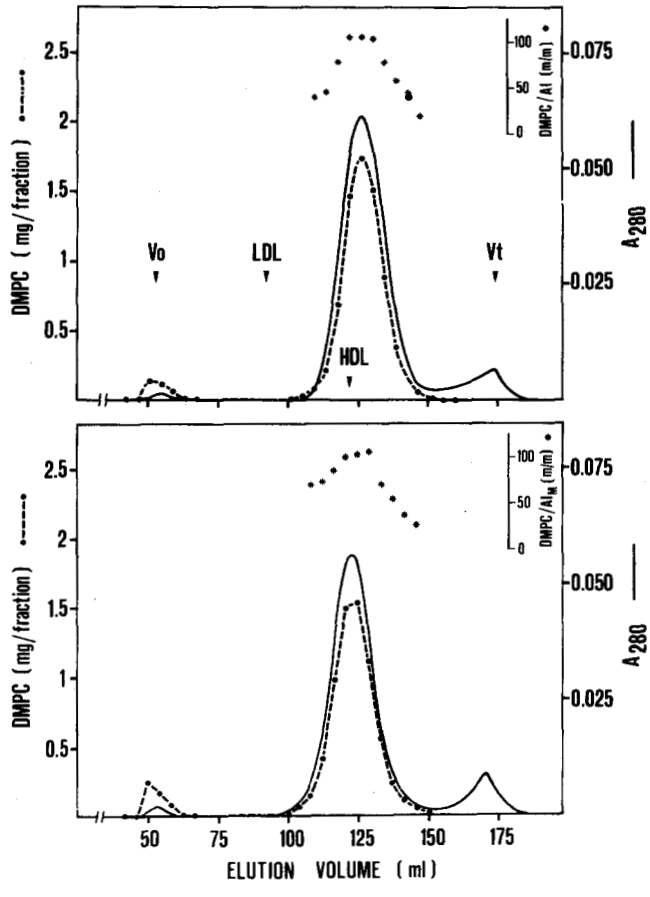


FIG. 1. **Agarose A-5m column chromatography of DMPC-AI (100:1 mol/mol) incubation mixtures.** Equal aliquots of DMPC were incubated at 25 °C for 20 h with either 2 mg of apo-AI (*upper panel*) or 2 mg of apo-AI_M (*lower panel*). The incubation mixtures were applied to a column (1.5 × 90 cm) of Agarose A-5m and eluted with 0.01 M Tris-HCl, 0.15 M NaCl, 1 mM EDTA, pH 7.4. Fractions of 4 ml were collected for protein and phospholipid determinations. The void volume (V_0) and the total volume (V_t) of the column, as well as the elution position of LDL and HDL, are indicated by *arrows*.

TABLE I

Characteristics of the complexes of normal apo-AI and variant apo-AI_{Milano} with DMPC

	AI	AI_{Milano}
Original mixture		
mol DMPC/mol AI	100	100
Complexes isolated by gel filtration		
Elution volume (ml)	125.9 ± 2.1[a]	124.0 ± 3.2
r (Å)	51.6 ± 1.3	51.0 ± 1.9
mol DMPC/mol AI	95.4 ± 4.7	94.3 ± 4.2
Fluorescence wavelength maximum (nm)	333.0[b]	334.7
Relative fluorescence intensity	128.0[b]	127.0
% α-helix	75.0	61.0

[a] Mean ± S.D. of four experiments.
[b] Mean of two experiments.

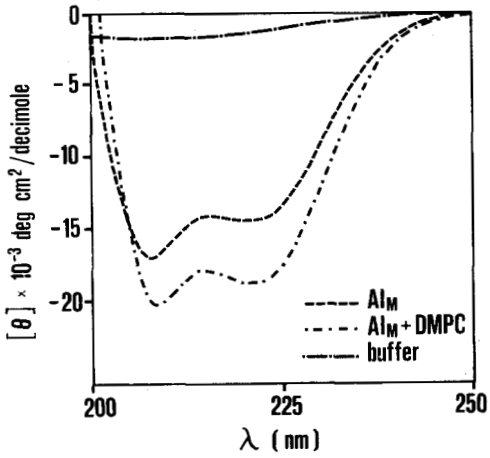


FIG. 2. **Far-ultraviolet CD spectra of apo-AI_M and of apo-AI_M-DMPC complexes.** The three central fractions of the Agarose A-5m peak were concentrated with an Amicon cell, equipped with a PM10 membrane, and dialyzed against 0.02 M phosphate buffer, pH 7.4, before the spectra were recorded.

content of apo-AI from 46% to 75% upon interaction with DMPC; a similar, but less pronounced increase was noted for apo-AI_{Milano} (from 39% to 61%) (Fig. 2).

Kinetics of Dimyristoylphosphatidylcholine Interaction with Apo-AI and Apo-AI_{Milano}—The association kinetics of DMPC with apo-AI and apo-AI_{Milano} were followed by measuring the clearance of liposomal turbidity as a function of time. This system does not discriminate between apo-AI binding to the surface of lipid bilayer and lipid-protein complex formation (10, 31). The reported data represent, therefore, the comparative interaction of the two apolipoproteins with the DMPC (11, 13, 32). The temperature of 25 °C was selected because the apoprotein association with phosphatidylcholine is fastest near the transition temperature of the phospholipid (24 °C for DMPC) (13). The absorbance curves at 325 nm of DMPC-apoprotein mixtures (100:1 mol/mol) containing apo-AI or apo-AI_{Milano} (0.1 mg/ml) are reported in Fig. 3. As the curves do not clearly obey to simple kinetic behaviors, the data have been plotted according to first order kinetics, also with the objective of obtaining comparative data with previously published observations on the apo-AI-DMPC association (13). A rate constant was defined as described by Pownall *et al.* (11): $K_{½} = (1/t_{½})$, where $t_{½}$ is the time necessary for the 50% disappearance of the relative turbidity.

The association of native apo-AI_{Milano} with DMPC is faster than for normal apo-AI, the calculated $K_{½}$ being 0.17 min^{-1}

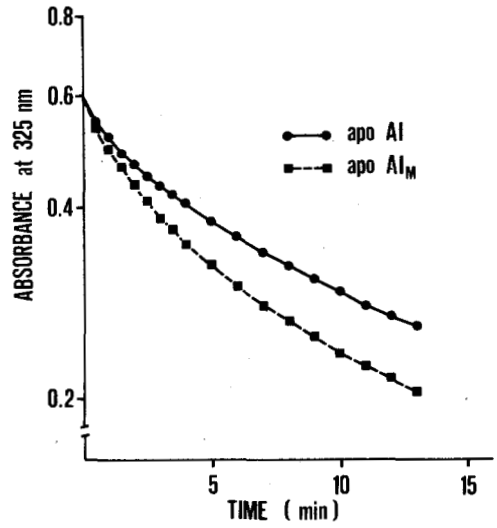


FIG. 3. **Association kinetics of apoproteins with DMPC.** The log of DMPC turbidity at 325 nm is plotted *versus* time, after DMPC liposomes and apo-AI or apo-AI_M (0.1 mg/ml) are combined at 25 °C with a 100:1 DMPC/AI molar ratio.

for apo-AI_{Milano} and 0.10 min^{-1} for apo-AI. The addition of increasing concentrations of GdnHCl to the incubation mixtures dramatically raises the rate constants for both apoproteins. The calculated $K_{½}$ for apo-AI_{Milano} and apo-AI were, respectively, 0.41 min^{-1} and 0.20 min^{-1} at GdnHCl = 0.25 M, increasing to 0.56 min^{-1} and 0.28 min^{-1} at GdnHCl = 0.5 M.

Stability of Isolated Complexes—To evaluate the stability of purified DMPC-AI and DMPC-AI_{Milano} complexes, we compared their behavior during exposure to increasing concentrations of GdnHCl, which is known to remove apo-AI both from artificial lipid-protein complexes (14, 27) and from native HDL (33). Fluorescence emission spectra of isolated complexes (0.05 mg of protein/ml) were recorded between 250 and 500 nm using an excitation wavelength of 280 nm. The addition of GdnHCl to the complexes induced a progressive shift of the wavelength maxima to higher values; concomitantly, the recorded fluorescence intensity at the wavelength maximum decreased about 30%. The denaturation processes of complexes containing apo-AI and apo-AI_{Milano} were monitored by plotting the relative fluorescence at the wavelength maxima against the GdnHCl concentration (Fig. 4). The denaturation pattern is similar for both DMPC-AI and DMPC-AI_{Milano}, the midpoint of the denaturation curve for DMPC-AI_{Milano} lying, however, at lower GdnHCl concentrations (3.4 M *versus* 4.0 M). This difference has been constantly observed with several preparations of DMPC-apoprotein complexes, suggesting a more labile structure for complexes with apo-AI_{Milano} than with apo-AI.

Physicochemical Properties of Apo-AI_{Milano}—To better evaluate some differences in the physicochemical behavior of apo-AI_{Milano} and apo-AI, emerging from the recombination studies, the apoproteins used in the described experiments and two other preparations of apo-AI and apo-AI_{Milano} were compared by spectrofluoroscopy and circular dichroism. The emission spectra of apo-AI_{Milano} excited at 280 nm were recorded between 250 and 500 nm, using a protein concentration of 0.05 mg/ml. The calculated fluorescence peak wavelength recorded in two analyses for each different preparation was of 337.4 ± 0.7 nm. When apo-AI was analyzed in the same conditions, a slightly lower peak wavelength was recorded: 335.2 ± 0.7 nm ($p < 0.005$).

The stability of the two apoproteins against denaturation

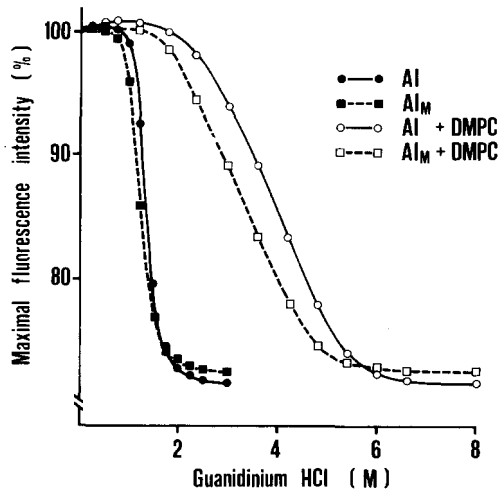


FIG. 4. **Influence of the incubation of apo-AI, apo-AI_M, and their complexes with DMPC with increasing concentrations of GdnHCl on the tryptophanyl fluorescence emission.** Purified apoproteins and complexes (0.05 mg of protein/ml) were incubated for 4 h at 25 °C with GdnHCl before the spectra were recorded.

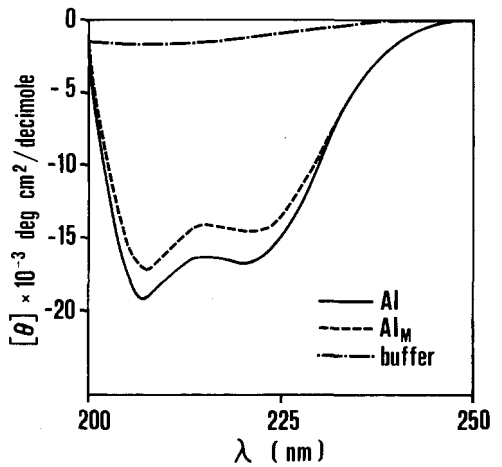


FIG. 5. **Far-ultraviolet CD spectra of purified apo-AI and apo-AI_M.** Apoproteins (0.1 mg/ml) were dialyzed against 20 mM phosphate buffer, pH 7.4, before spectra were recorded.

by GdnHCl was evaluated by spectrofluoroscopy, as described for the DMPC-apoprotein complexes (Fig. 4). Both apoproteins give a clear red shift at a GdnHCl concentration between 1 and 2 M, the denaturation curve being slightly shifted to lower GdnHCl concentrations for apo-AI_{Milano}.

Secondary structures of apo-AI and apo-AI_{Milano} were studied in terms of far-ultraviolet circular dichroism spectra. To prevent possible self-associations of the apoproteins in buffer, which can modify their circular dichroism spectra (34), low protein concentrations (0.10 mg/ml) were used in all the experiments. Both apoproteins give a quite similar profile (Fig. 5), indicative of a high content of α-helix; however, the minima at 208 and 222 nm are more pronounced with normal apo-AI. The calculated percentage of α-helical content in apo-AI_{Milano} is slightly lower compared to normal apo-AI (AI_{Milano} = 39%, 40%, 37%; AI = 46%, 57%, 59%).

DISCUSSION

DMPC has been widely used to characterize the lipid binding properties of different structurally normal (AI, AII, CI, CII, CIII, E) and abnormal ($AI_{Tangier}$) apolipoproteins (6, 35). In the present study, we used DMPC to investigate the interaction of lipids with the apoprotein AI_{Milano}, a mutant, characterized by the $Arg^{173} \rightarrow$ Cys substitution.

Apo-AI_{Milano} readily associates with DMPC to form stable complexes, not differing in size from those obtained in recombination experiments with normal apo-AI. As the association of apo-AI with DMPC is markedly affected by temperature (13) and by proportions of reactants (4, 10), all recombination experiments were carried out at 25 °C, where the rate of association is maximal (13), and with a molar ratio DMPC/apo-AI of 100:1, at which a nearly total incorporation of apo-AI in a small lipid-protein micellar complex can be obtained (4). In these conditions, both apo-AI and apo-AI_{Milano} are incorporated to the same extent in stable complexes clearly separable by gel filtration and showing nearly identical dimensions and stoichiometries.

Although the same end product has been obtained with the two apolipoproteins, the DMPC-AI_{Milano} complexes denature at lower GdnHCl concentrations, compared to DMPC-AI, suggesting that apo-AI_{Milano} is less tightly associated to lipids than normal apo-AI. On the other hand, a higher affinity of apo-AI_{Milano} for DMPC was suggested by the 50% higher association rate of apo-AI_{Milano}, compared to normal apo-AI.

The association of apolipoproteins with phospholipids is apparently kinetically controlled, the kinetic behavior being dependent upon different factors: the association rate is fastest near the transition temperature of the phospholipid (13), the rate of association with DMPC of different apolipoproteins increases with decreasing molecular weight of the apoprotein (11), a higher number of exposed hydrophobic residues in the apolipoprotein increases the association rate (11). The accelerated kinetic behavior of apo-AI_{Milano} compared to apo-AI cannot be explained by differences in the phospholipid matrix, as similar reaction conditions (temperature, molar ratios) were employed in all the recombination procedures.

Structural differences between the two apolipoproteins are probably responsible for the faster association rate observed with apo-AI_{Milano}. The variant apoprotein can form high molecular weight dimers (20), but remains monomeric during the recombination experiments as assessed by SDS-polyacrylamide gel electrophoresis of delipidated DMPC-AI_{Milano} complexes in nonreducing conditions (data not shown), thus excluding a molecular weight effect. All kinetic experiments were carried out at low protein concentration (0.1 mg/ml), thus preventing the possible formation of apolipoprotein aggregates (34); it cannot be excluded that differences in the self-association behavior between the two apolipoproteins may somehow be responsible for the different kinetics. However, kinetic analyses carried out in the presence of 0.5 M GdnHCl, where apo-AI is only monomeric (36), constantly show a higher association rate for the apo-AI_{Milano}.

In order to evaluate which other differences in the secondary structure between apo-AI_{Milano} and apo-AI are responsible for the faster association rate of the variant, the two apolipoproteins were analyzed by spectrofluoroscopy and circular dichroism. The intrinsic fluorescence spectra of different preparations of apo-AI_{Milano} constantly show a higher peak wavelength compared to normal apo-AI. Although the difference is minimal, it is extremely reproducible and statistically significant. When the secondary structures of the two apolipoproteins were examined by circular dichroism, a lower content of α-helix could be demonstrated in the apo-AI_{Milano}. These findings suggest that the conformation of apo-AI_{Milano} is slightly different from that of normal apo-AI, leading to an increased exposition to the solvent of hydrophobic residues in the variant apolipoprotein.

A support to this hypothesis comes from the observation of

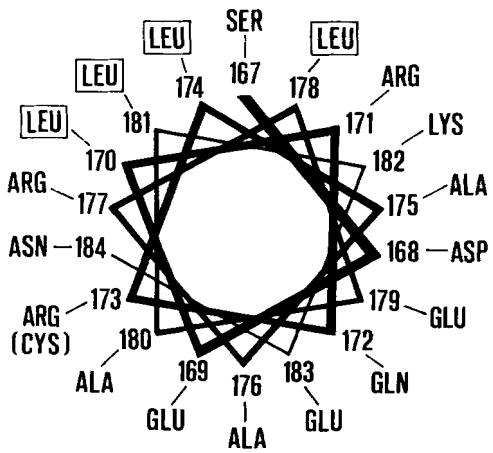


FIG. 6. **Helical wheel presentation of the presumed α-helical structure of segment 167–184 of apo-AI. In apo-AI_M, the Arg^{173} is substituted with cysteine.** The *boxed residues* mark the hydrophobic face of the amphipathic helix.

an anomalous distribution of charged residues in the apolipoprotein fragment where the Arg → Cys substitution takes place. The amino acid sequence of the apo-AI fragment 167–184 is reported in Fig. 6, according to the Edmunson model of helical wheels (37). This fragment is part of one (segment 166–186) of the several regions in apo-AI presumed to be α-helices (38). In particular, in this amphipathic fragment, the hydrophobic face is occupied by 4 Leu residues (*boxed* in Fig. 6), whereas the negatively charged residues (Asp^{168}, Glu^{169}, Glu^{179}, Glu^{183}) are located on the opposite side. The positively charged residues (Arg^{171}, Arg^{173}, Arg^{177}, Lys^{182}) occur at the interface between the polar and apolar faces of the helix. Two ion pairs (39) in topographically close relationship are present in this fragment of apo-AI: $Asp^{168} - Arg^{171}$ and $Glu^{169} - Arg^{173}$. The substitution of Arg^{173} with Cys leads to the disappearance of one of these ion pairs, with a resulting loss in amphipathic structure and lipid binding capacity (40). As a consequence, the total secondary structure in the apo-AI_{Milano} may be remodeled, resulting in a decreased total α-helical content and, possibly, in a slightly higher exposure of hydrophobic residues. The association with phospholipids will proceed at a faster rate (11) and since the number of helical segments in apo-AI_{Milano} involved in lipid interaction is lower than for apo-AI, the former will be more easily displaced from the complexes.

The physiological significance of the reported findings should be considered, in the light of the described abnormalities of lipoprotein metabolism in the AI_{Milano} carriers, as well as taking into account the heterogeneity of the HDL particles in these (23), possibly reflecting differences in the molecular forms of the variant apo-AI (20). Further studies will investigate the lipid binding properties of the disulfide-linked dimers and complexes with apo-AII (20). The reported results, however, clearly indicate significant differences in the dynamic interaction with lipids of the mutant apolipoprotein, compared to normal AI. The higher kinetic affinity of apo-AI_{Milano} for lipids and, on the other hand, the easier desorption of apo-AI_{Milano} from lipid-protein complexes, may contribute to its accelerated catabolism (41), as well as to an efficient uptake capacity for tissue lipid.

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